Methods Section Draft

Chaotic Hamiltonian Dynamics of Polyribosome Traffic on Circular mRNA: Beyond TASEP

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1 Methods

1.1 Model Formulation

1.1.1 System Definition and Physical Framework

We model a polyribosome complex as N hard spheres constrained to a one-dimensional circular track of length L, representing a circularized mRNA strand. The circular topology reflects the well-characterized closed-loop conformation adopted by eukaryotic mRNAs through eIF4E-eIF4G-PABP bridging interactions [**?**, **?**]. This geometry enables Closed-Loop-Assisted Reinitiation (CLAR), wherein terminating ribosomes are efficiently recycled to the start codon without diffusing into the bulk cytoplasm [**?**].

Each ribosome i is characterized by a time-dependent state vector:

$$\mathbf{s}_i(t) = (x_i(t),\, v_i(t),\, d_i(t),\, m_i(t)) \tag{1}$$

where $x_i \in [0, L)$ is the position along the mRNA (in codon units), v_i is the translocation velocity, d_i is the effective collision diameter, and m_i is the effective mass. Crucially, both $d_i(t)$ and $m_i(t)$ evolve as the nascent polypeptide chain grows during translation, distinguishing our model from classical hard-sphere systems with fixed particle properties.

The system obeys strict single-file diffusion (SFD) constraints: ribosomes cannot pass one another due to their physical size relative to the mRNA track width. This ordering constraint is naturally satisfied by the hard-sphere collision dynamics.

1.1.2 Hamiltonian Dynamics

The system evolves according to deterministic Hamiltonian mechanics between collisions. The Hamiltonian is:

$$\mathcal{H} = \sum_{i=1}^{N} \frac{p_i^2}{2m_i(t)} + \sum_{i<j} V_{\text{HS}}(r_{ij}) \tag{2}$$

where $p_i = m_i v_i$ is the momentum of ribosome i, and V_{HS} is the hard-sphere potential:

$$V_{\text{HS}}(r_{ij}) = \begin{cases} \infty & \text{if } r_{ij} < \frac{1}{2}(d_i + d_j) \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

Here, $r_{ij} = \min(|x_i - x_j|, L - |x_i - x_j|)$ is the geodesic distance on the circular track.

Between collisions, ribosomes undergo free translocation:

$$\frac{dx_i}{dt} = v_i, \quad \frac{dp_i}{dt} = 0 \tag{4}$$

At the moment of collision between ribosomes i and j, velocities are updated according to the one-dimensional elastic collision formulas:

$$v_i' = \frac{(m_i - m_j)v_i + 2m_j v_j}{m_i + m_j} \tag{5}$$

$$v_j' = \frac{(m_j - m_i)v_j + 2m_i v_i}{m_i + m_j} \tag{6}$$

These equations conserve both total momentum $P = \sum_i m_i v_i$ and total kinetic energy $E = \sum_i \frac{1}{2} m_i v_i^2$.

1.1.3 Nascent Chain Dynamics and Mass/Diameter Evolution

The key biological innovation of our model is the explicit coupling between translational progress and ribosome physical properties. As ribosome i translates, its nascent polypeptide chain grows at a rate proportional to the translocation velocity:

$$\frac{d\ell_i}{dt} = v_i \tag{7}$$

where $\ell_i(t)$ is the chain length in amino acid residues (equivalent to codons translated).

The nascent chain contributes to the effective mass:

$$m_i(t) = m_{\text{ribo}} + \ell_i(t) \cdot m_{\text{aa}} \tag{8}$$

where $m_{\text{ribo}} = 4.2$ MDa is the 80S ribosome mass and $m_{\text{aa}} \approx 110$ Da is the average amino acid mass.

The effective diameter depends on both the ribosome footprint and the hydrodynamic radius of the emerged nascent chain. We model the chain as a polymer that may adopt different conformational states:

$$d_i(t) = d_{\text{ribo}} + 2R_g(\ell_i^{\text{emerged}}) \tag{9}$$

where $d_{\text{ribo}} \approx 10$ codons is the ribosome footprint, and R_g is the radius of gyration of the emerged portion of the nascent chain.

The emerged chain length accounts for the ribosomal exit tunnel:

$$\ell_i^{\text{emerged}} = \max\left(0,\, \ell_i - \ell_{\text{tunnel}}\right) \tag{10}$$

where $\ell_{\text{tunnel}} \approx 40$ residues is the exit tunnel capacity [**?**].

The radius of gyration follows polymer scaling laws dependent on the folding state:

$$R_g(N) = b \cdot N^{\nu} \tag{11}$$

where $b \approx 1$ nm is the Kuhn length and the scaling exponent ν varies:

$$\nu = \begin{cases} 0.6 & \text{extended/random coil} \\ 0.5 & \text{partially folded} \\ 0.33 & \text{compact/globular} \end{cases} \tag{12}$$

We assign folding states based on emerged chain length, reflecting the progressive co-translational folding observed experimentally [**?**, **?**]:

$$\text{Folding state} = \begin{cases} \text{extended} & \ell^{\text{emerged}} < 50 \\ \text{partial} & 50 \leq \ell^{\text{emerged}} < 150 \\ \text{compact} & \ell^{\text{emerged}} \geq 150 \end{cases} \tag{13}$$

1.1.4 Circular mRNA Topology and Boundary Conditions

The mRNA is modeled as a circular track with distinct functional regions:

$$L = L_{5'\text{UTR}} + L_{\text{CDS}} + L_{3'\text{UTR}} \tag{14}$$

Positions are defined modulo L, implementing periodic boundary conditions:

$$x_i \mapsto x_i \mod L \tag{15}$$

Translation initiation occurs stochastically at the start codon position $x_{\text{start}} = L_{5'\text{UTR}}$ with rate k_{init}:

$$P(\text{initiation in } dt) = k_{\text{init}} \cdot dt \cdot \mathbf{1}_{\text{site clear}} \tag{16}$$

where $\mathbf{1}_{\text{site clear}}$ is an indicator function that equals 1 only if no ribosome occupies the initiation site.

Upon reaching the stop codon at $x_{\text{stop}} = L_{5'\text{UTR}} + L_{\text{CDS}}$, a ribosome terminates and, under CLAR, is recycled to the start codon:

$$x_i \to x_{\text{start}}, \quad \ell_i \to 0, \quad v_i \to v_0 \tag{17}$$

where v_0 is the baseline elongation rate.

1.2 Emergence of Chaos

1.2.1 Lyapunov Exponent and Predictability Horizon

For $N \geq 3$ particles with unequal masses, the system exhibits deterministic chaos characterized by a positive maximal Lyapunov exponent $\lambda_{\max} > 0$. This exponent quantifies the exponential divergence of initially nearby trajectories:

$$\|\delta\mathbf{x}(t)\| \sim \|\delta\mathbf{x}(0)\| \cdot e^{\lambda_{\max} t} \tag{18}$$

The Lyapunov time, $\tau_{\text{Lyap}} = 1/\lambda_{\max}$, defines the predictability horizon beyond which trajectory-level predictions become unreliable. We estimate $\lambda_{\max}$ numerically by evolving pairs of trajectories with infinitesimal initial separation.

1.2.2 Conditions for Chaos

The transition from integrable to chaotic dynamics depends critically on mass heterogeneity. For identical masses ($m_i = m$ for all i), the system is completely integrable: collisions simply exchange velocities, and the dynamics are equivalent to non-interacting particles passing through each other. This corresponds to $\lambda_{\max} = 0$.

Mass heterogeneity breaks integrability. We quantify heterogeneity via the coefficient of variation:

$$\text{CV}_m = \frac{\sigma_m}{\bar{m}} = \frac{\sqrt{\frac{1}{N}\sum_i (m_i - \bar{m})^2}}{\frac{1}{N}\sum_i m_i} \tag{19}$$

In polyribosome systems, mass heterogeneity arises naturally from the different nascent chain lengths, with CV_m evolving dynamically as translation proceeds.

1.3 Density Field Evolution and the Perron-Frobenius Operator

1.3.1 Coarse-Grained Density

While individual trajectories become unpredictable beyond τ_{Lyap}, the evolution of probability densities remains well-defined. We define the coarse-grained ribosome density field:

$$\rho(x,t) = \sum_{i=1}^{N} K_\sigma(x - x_i(t)) \tag{20}$$

where K_σ is a Gaussian kernel with width σ:

$$K_\sigma(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{x^2}{2\sigma^2}\right) \tag{21}$$

This density field is directly comparable to ribosome profiling (Ribo-seq) data, which measures ribosome occupancy along mRNA with codon-level resolution [**?**].

1.3.2 Perron-Frobenius Operator

The time evolution of the density field is governed by the Perron-Frobenius (PF) operator $\mathcal{P}^t$:

$$\rho(x, t + \Delta t) = \mathcal{P}^{\Delta t}\rho(x,t) \tag{22}$$

For a deterministic dynamical system with flow map Φ^t, the PF operator acts on densities via:

$$(\mathcal{P}^t\rho)(x) = \int \delta\left(x - \Phi^t(y)\right)\rho(y)\,dy \tag{23}$$

Crucially, the PF operator is *linear* even when the underlying dynamics are nonlinear and chaotic. This linearity enables systematic analysis and, as we exploit, machine learning approximation.

1.3.3 Neural Operator Approximation

We approximate the PF operator using a Fourier Neural Operator (FNO) architecture [**?**]. The FNO learns the mapping:

$$\rho(x,t) \mapsto \rho(x, t + \Delta t) \tag{24}$$

directly from simulation data, without requiring explicit knowledge of the underlying equations.

The FNO architecture consists of:

1. A lifting layer: $\mathbb{R} \to \mathbb{R}^{d_h}$ (density to hidden channels)
2. L Fourier layers with spectral convolutions
3. A projection layer: $\mathbb{R}^{d_h} \to \mathbb{R}$ (hidden channels to density)

Each Fourier layer performs:

$$v^{(l+1)}(x) = \sigma\left(W^{(l)}v^{(l)}(x) + \mathcal{F}^{-1}\left(R^{(l)} \cdot \mathcal{F}(v^{(l)})\right)(x)\right) \tag{25}$$

where $\mathcal{F}$ denotes the Fourier transform, $R^{(l)}$ are learnable spectral weights, and σ is a nonlinear activation (GeLU).

The periodic boundary conditions of the circular track make the Fourier basis natural, as it automatically respects the topology.

1.4 Comparison with TASEP

1.4.1 Totally Asymmetric Simple Exclusion Process

The standard model for ribosome traffic is the Totally Asymmetric Simple Exclusion Process (TASEP) [**?**, **?**]. TASEP is a discrete-time, discrete-space stochastic model where:

- The mRNA is discretized into L sites
- Each site is either empty or occupied by one ribosome
- At each timestep, a ribosome hops forward with probability p if the next site is empty
- Ribosomes enter at the first site with rate α and exit from the last with rate β

For a periodic (circular) track with homogeneous hopping rate p, the steady-state density is uniform:

$$\rho_{\text{TASEP}}^{\text{steady}} = \frac{N}{L} \tag{26}$$

1.4.2 Key Differences from Our Model

Our Hamiltonian model differs from TASEP in several fundamental ways:

Property	**TASEP**	**Hamiltonian Model**
Dynamics	Stochastic, Markovian	Deterministic, chaotic
Time	Discrete	Continuous
Space	Discrete lattice	Continuous
Particle properties	Identical	Heterogeneous, time-varying
Collisions	Exclusion (blocking)	Elastic momentum transfer
Memory	Memoryless	Full phase-space memory

Table 1: Comparison of TASEP and Hamiltonian polyribosome models.

These differences lead to distinct predictions for:

1. **Fluctuation statistics**: TASEP predicts Poissonian fluctuations; our model predicts correlated fluctuations arising from deterministic chaos
2. **Density correlations**: TASEP has short-range correlations; chaotic dynamics can produce long-range correlations
3. **Traffic jam dynamics**: TASEP jams dissolve stochastically; Hamiltonian jams exhibit inertial effects

1.5 Simulation Algorithm

1.5.1 Event-Driven Molecular Dynamics

We employ an event-driven simulation algorithm that computes exact collision times rather than using fixed timestep integration. This approach:

- Eliminates discretization error in collision detection
- Provides exact collision timestamps for machine learning
- Is efficient when collision frequency is moderate

The algorithm proceeds as follows:

1. **Initialize**: Place N ribosomes with initial positions, velocities, and nascent chain lengths

2. **Compute collision times**: For each adjacent pair $(i, i+1)$ on the circular track, compute:

$$t_{ij}^{\text{collision}} = \frac{r_{ij} - \frac{1}{2}(d_i + d_j)}{v_i - v_j} \tag{27}$$

   retaining only positive, finite values
3. **Find next event**: $t^* = \min_{\{i,j\}} t_{ij}^{\text{collision}}$
4. **Advance system**: Update all positions: $x_i \to x_i + v_i \cdot t^*$
5. **Update nascent chains**: $\ell_i \to \ell_i + v_i \cdot t^*$, then recompute m_i and d_i
6. **Resolve collision**: Apply elastic collision formulas to colliding pair
7. **Handle termination**: If any ribosome crosses the stop codon, apply CLAR recycling
8. **Handle initiation**: With probability $k_{\text{init}} \cdot t^*$, attempt to add a new ribosome
9. **Repeat** from step 2 until target simulation time

1.5.2 Conservation Verification

At each timestep, we verify conservation laws:

$$|E(t) - E(0)| < \epsilon_E \tag{28}$$

$$|P(t) - P(0)| < \epsilon_P \tag{29}$$

where $\epsilon_E, \epsilon_P \sim 10^{-10}$ are numerical tolerance thresholds. Violations indicate algorithmic errors.

1.6 Biological Parameters

We parameterize the model using experimentally measured values from the literature (Table **??**).

Parameter	**Symbol**	**Value**	**Reference**
Ribosome mass	m_{ribo}	4.2 MDa	[**?**]
Ribosome footprint	d_{ribo}	28–30 nt (≈10 codons)	[**?**]
Exit tunnel length	ℓ_{tunnel}	40 residues	[**?**]
Amino acid mass	m_{aa}	110 Da (average)	–
Elongation rate	v_0	5–10 codons/s	[**?**]
Initiation rate	k_{init}	0.03–0.3 s^{-1}	[**?**]
Typical CDS length	L_{CDS}	300–500 codons	–

Table 2: Biological parameters used in simulations.

1.7 Data Analysis and Validation

1.7.1 Comparison with Ribosome Profiling Data

Ribosome profiling (Ribo-seq) provides genome-wide measurements of ribosome positions along mRNAs [**?**]. The technique generates sequencing reads proportional to ribosome density, enabling direct comparison with our simulated density fields.

For validation, we:

1. Obtain published Ribo-seq datasets for well-characterized mRNAs

2. Compute the time-averaged density profile from simulations:

$$\bar{\rho}(x) = \frac{1}{T} \int_0^T \rho(x,t)\, dt \tag{30}$$

3. Compare with normalized Ribo-seq read counts using Pearson correlation and root-mean-square error

1.7.2 Statistical Measures

We characterize the system using:

- **Ribosome density**: $\langle N \rangle / L_{\text{CDS}}$
- **Translation rate**: Number of completed translations per unit time
- **Mass heterogeneity**: $\text{CV}_m(t)$ as a function of time
- **Collision frequency**: Collisions per ribosome per unit time
- **Traffic jam duration**: Time intervals where local density exceeds a threshold

1.7.3 Lyapunov Exponent Estimation

We estimate $\lambda_{\max}$ using the standard algorithm:

1. Initialize two trajectories with separation $\|\delta \mathbf{x}(0)\| = \epsilon_0$
2. Evolve both for time τ
3. Record $\|\delta \mathbf{x}(\tau)\|$
4. Rescale the perturbed trajectory to restore separation ϵ_0
5. Repeat and average:

$$\lambda_{\max} \approx \frac{1}{n\tau} \sum_{k=1}^{n} \ln \frac{\|\delta \mathbf{x}(k\tau)\|}{\epsilon_0} \tag{31}$$

1.8 Software Implementation

The simulation code is implemented in Python with the following components:

- `core.ParticleSystem`: Base hard-sphere dynamics engine
- `biology.Ribosome`: Ribosome with nascent chain growth
- `biology.PolyribosomeSystem`: Full polyribosome simulation
- `neural_operators.FourierNeuralOperator`: FNO for density prediction

Neural operator training uses PyTorch with the Adam optimizer. Simulations were performed on [hardware details]. Code is available at [repository URL].

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